

ORIGINAL ARTICLE

Analysis of *fks1* and *fks2* gene mutations in invasive *Candida glabrata* strains from Pakistan

Saba Memon^{1,2} | Najia Karim Ghanchi¹ | Urooj Zafar² | Joveria Farooqi¹ | Sadaf Zaka¹ | Kauser Jabeen¹ 

¹Department of Pathology & Laboratory Medicine, Aga Khan University, Karachi, Pakistan

²Department of Microbiology, University of Karachi, Karachi, Pakistan

Correspondence

Kauser Jabeen, Department of Pathology & Laboratory Medicine, The Aga Khan University, Stadium Road, Karachi-74800, Pakistan.

Email: kausar.jabeen@aku.edu

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Abstract

Introduction: The gradual increase in caspofungin usage in Pakistan raises a concern of emergence of echinocandin resistance in local *Candida glabrata* strains. We sequenced and determined mutations in *fks1* and *fks2* genes in invasive *Candida glabrata* strains from Pakistan.

Material and Methods: Thirty-six invasive *C. glabrata* strains were selected with median (min-max) minimum inhibitory concentrations (MICs) of 0.06 (0.015–0.25) mg/L for caspofungin, 0.015 (0.008–0.06) mg/L for micafungin and 0.06 (0.015–0.12) mg/L for anidulafungin. *fks1* and *fks2* gene fragments were sequenced using Sanger methodology. Sequences were analysed with MEGA-6 software to identify specific single-nucleotide polymorphisms (SNP) against wild-type sequences of *C. glabrata*.

Results: In *fks1* gene, non-synonymous mutation D632H was observed in one isolate with caspofungin MIC of 0.25 mg/L. Synonymous mutation at position A742 was observed in 26/36 (72%) of the isolates. 34/36 (94.5%) isolates analysed for *fks2* gene were observed as wild type. A novel non-synonymous mutation at I661T was observed in *fks2* gene in one isolate with caspofungin MIC of 0.12 mg/L and anidulafungin and micafungin MIC of 0.06 and 0.015 mg/L, respectively. Novel *fks2* synonymous mutations at position T647, K652 and I706 were observed in 16/36 (44%), 25/36 (69%) and 23/36 (63%) isolates, respectively.

Conclusion: Low frequencies of both non-synonymous and synonymous polymorphisms were observed in invasive *C. glabrata* strains. Since S663P in *fks2* gene is associated with caspofungin resistance, a novel mutation at 661 codon identified in our study needs correlation with treatment outcome data and mandates periodic genomic surveillance.

KEYWORDS

Candida glabrata, echinocandin resistance, *fks*

1 | INTRODUCTION

Infections due to *Candida glabrata* are of global concern as *C. glabrata* exhibits greater rates of antifungal resistance than other *Candida* species.¹ The frequency of invasive *C. glabrata* reported

from tertiary care centres ranges between 5% and 10.7% in Pakistan and 18.0% in India.^{2–4}

Echinocandins are the first line antifungal therapy for *C. glabrata* infections because of their better efficacy and broader antifungal spectrum.⁵ Therefore, reports of emerging resistance

to echinocandins in *C. glabrata* pose a serious clinical challenge. Increasing trends of echinocandin resistance (around 3% in 2008 to around 10% in 2013) in *C. glabrata* was reported from four US cities.⁶ Similarly, a nation-wide fungemia survey from Denmark reported an increase in echinocandin resistance in *C. glabrata* from 0.9% in 2008–2009 to 3.1% in 2012–2013.

In contrast, data from the UK, Germany, Austria, India, Iran and Turkey reported low to negligible rates of resistance.^{7–11} Echinocandin resistance has also been associated with cross-resistance to azole in 36% of the echinocandin-resistant strains, emphasising the issue of multidrug-resistant (MDR) *C. glabrata*.⁶ Caspofungin use has been gradually increasing in Pakistan over the past few years, raising the possibility of emergence of echinocandin resistance in local clinical strains.¹²

Echinocandin targets the biosynthesis of fungal cell wall by inhibiting β -1,3-D-glucan synthase enzyme synthesis encoded by *FKS* gene. Reduced susceptibility to echinocandins is associated with amino acid substitutions in two highly conserved hotspot regions (HS1 and HS2) of the *FKS1* and *FKS2* genes.¹³ These mutations are two times more common in *FKS2* when compared with *FKS1* in *C. glabrata*.¹⁴

Additionally several studies have shown that the presence of non-synonymous mutations in hotspots 1 and 2 of *FKS1* and *FKS2* region is the most important independent risk factor in predicting echinocandin therapeutic responses among patients with invasive candidiasis.¹⁵

Commonly reported single-nucleotide polymorphism associated with echinocandin resistance in *FKS2* gene are at positions D666E/H/N, Δ F659, L664R and S663F/P^{16–19} S629P and D632E mutation in *FKS1* gene.^{18,20,21}

Echinocandin resistance is defined if an isolate demonstrates higher minimum inhibitory concentration (MIC) to at least two echinocandins.²² As high caspofungin MICs alone do not confirm echinocandin resistance due to excessive inter-laboratory variability,²³ both CLSI and EUCAST recommend micafungin or anidulafungin as surrogate markers for caspofungin susceptibility testing. Phenotypic and genotypic profiling of echinocandin resistance may provide reliable data on emergence of antifungal resistance in Pakistan. In this study, we present frequencies of *FKS1* and *FKS2* gene mutations in *C. glabrata* isolates from Pakistan.

2 | MATERIALS AND METHODS

A retrospective cross-sectional study was conducted at the Aga Khan University clinical laboratories, Karachi, Pakistan. The study was exempted from full review by the institute's Ethics Review Committee (AKU-ERC number: 2019-0438-2659).

2.1 | Selection of isolates

According to Pfaller et al.,²⁴ although isolates with *FKS* mutations were more likely to have higher MICs of caspofungin, about one fourth (14/60) of the isolates with *FKS* mutations had low caspofungin

MICs. Hence, 36 clinical isolates from various sources were selected from our previous study across the full range of MICs obtained against caspofungin for detection of *FKS* mutations from 0.03 to 0.25 mg/L (Table 1).²⁵

To ensure geographical representation from across Pakistan, 30 patients were from Karachi (25 from Aga Khan University Hospital), three from other areas of Sindh province, two from Punjab and one from Baluchistan. We also aimed to represent different levels of health care services, so the 25 registered patients were from emergency department, critical care, non-critical hospitalised patients, and out-patients.

2.2 | Phenotypic identification

The identification of the isolates was confirmed through standard identification protocols.²⁶ Isolates were identified by type of growth on BBL™ CHROMagar™ Candida (BD, USA) and Bismuth Glucose Glycine Yeast Agar (Oxoid, UK), Dalmat method, i.e., microscopy on Cornmeal agar with Tween80 (Oxoid, UK). API 20C Aux (bioMérieux, France) was used to confirm identification through biochemical profile at 48–72 h.

2.3 | Antifungal susceptibility

MICs were performed on YeastOne Sensititre (Trek Diagnostics System Ltd.) using colorimetric broth microdilution with Roswell Park Memorial Institute (RPMI) 1640 broth (Sigma Aldrich, United Kingdom) as described previously.²⁵ Antifungal susceptibility testing was performed on fresh isolated colonies of *C. glabrata* that transferred into 5 ml deionised water and matched with 0.5 McFarland standard. 20 μ l of the suspension was transferred to 11 ml of RPMI broth and 100 μ l of that was inoculated into each well of the Sensititre plate. For control, 10 μ l from the first well was inoculated into 5% Sheep Blood Agar plate and incubated at 35°C for 24 h. *C. parapsilosis* ATCC 22019 and *C. krusei* ATCC 6258 were used as controls. Colour change from blue to pink was considered positive for growth in that well and MICs were thus recorded.

Susceptibilities were interpreted according to Clinical Laboratory Standards Institute (CLSI) breakpoints mentioned in "Performance standard for antifungal susceptibility testing of yeasts M60-ED: 2017". The susceptibility of the isolates was interpreted as follows: anidulafungin and caspofungin MIC $\leq$ 0.12 mg/L was susceptible, MIC = 0.25 mg/L was intermediate and MIC $\geq$ 0.5 mg/L resistant. For micafungin, MIC $\leq$ 0.06 mg/L was susceptible, MIC = 0.12 mg/L intermediate and MIC $\geq$ 0.25 mg/L resistant.

2.4 | DNA extraction and Molecular identification

Fungal genomic DNA was extracted by using Qiagen mini kit (QIAquick, Germany) protocol with minor modification. Fungal

TABLE 1 MICs and *fk*s non-synonymous and synonymous mutation in *Candida glabrata*

S. no.	Strain	Source	MIC (mg/L)		CAS-MIC	ANI- MIC	MCF-MIC	<i>fk</i> s mutations		Antifungal use	Outcome
			FLU-MIC	AMB-MIC				<i>fk</i> s1	<i>fk</i> s2		
1	AKU-2009-625	Blood	8	0.25	0.25	0.06	0.015	D632H	K652, Y653, Y682, I706	No	Alive
2	AKU-2011-800	Blood	64	0.5	0.12	0.06	0.015		T647, K652, I706	No	Dead
3	AKU-2012-1118	Blood	64	0.25	0.12	0.06	0.03	A742	T647, K652, I706	No	Alive
4	AKU-2012-1009	Blood	64	0.5	0.06	0.03	0.015	A742	K652, Y653, Y682, I706	FLU, AMB	Dead
5	AKU-2018-132	Blood	256	0.25	0.06	-	-	A742	T647, K652, I706	AMB	Alive
6	AKU-2019-134	Blood	2	0.25	0.03	0.03	0.015	A742	K652, Y653, Y682, I706	FLU, AMB	NA
7	AKU-2019-282	Blood	64	0.5	0.12	0.06	0.008			NA	NA
8	AKU-2019-334	Blood	64	0.5	0.06	0.06	0.015		T647, K652, I706	FLU, AMB	Alive
9	AKU-2019-460	Blood	16	0.25	0.015	0.015	0.015	A742	T647, K652, I706	NA	NA
10	AKU-2019-496	Blood	256	0.5	0.06	0.03	0.015	A742	T647, K652, I706	AMB, CAS	NA
11	AKU-2020-94	Blood	256	0.5	0.06	0.03	0.015	A742	K652, Y682	AMB	NA
12	AKU-2012-1269	Fluid	8	0.25	0.25	0.03	0.015	A742	W639	No	Dead
13	AKU-2019-35	Fluid	8	0.12	0.03	0.12	0.06	A742	T647, K652, I706	No	NA
14	AKU-2019-38	Fluid	64	0.12	0.12	0.015	0.008	A742		No	Alive
15	AKU-2019-165	Fluid	8	0.5	0.25	0.12	0.015		W639	No	Alive
16	AKU-2019-226	Fluid	2	0.25	0.06	0.03	0.015	A742	W639	AMB	NA
17	AKU-2019-302	Fluid	8	0.5	0.03	0.03	0.008		T647, K652, I706	NA	NA
18	AKU-2019-313	Fluid	256	0.12	0.12	0.06	0.015	A742	T647, K652, I706	No	Dead
19	AKU-2019-435	Fluid	16	0.25	0.06	0.06	0.015		T647, K652, I706	FLU, AMB	NA
20	AKU-2019-485	Fluid	64	0.5	0.06	0.06	0.015	A742	T647, K652, I706	FLU, VORI	NA
21	AKU-2020-193	Fluid	16	0.25	0.25	0.12	0.015	A742	T647, K652, I706	No	Alive
22	AKU-2011-809	Pus	32	0.12	0.25	0.06	0.015		K652, Y653, Y682, I706	FLU, AMB	Dead
23	AKU-2012-984	Pus	64	0.25	0.06	0.06	0.015	A742		FLU	Alive
24	AKU-2019-84	Pus	64	0.5	0.12	0.06	0.015		I661T, W639	NA	NA
25	AKU-2019-90	Pus	128	0.5	0.06	0.06	0.015		T647, K652, I706	NA	Dead
26	AKU-2019-197	Pus	64	0.25	0.06	0.03	0.008	A742	T647, K652, I706	NA	NA
27	AKU-2019-391	Pus	64	0.5	0.06	0.03	0.015	A742	K652, Y682	NA	NA
28	AKU-2019-438	Pus	16	0.5	0.06	0.03	0.015	A742	W639	FLU	NA
29	AKU-2020-45	Pus	64	0.5	0.12	0.06	0.015	A742	W639	NA	NA
30	AKU-2020-51	Pus	64	0.25	0.06	0.06	0.015	A742	K652, Y653, Y682, I706	NA	NA
31	AKU-2020-162	Pus	16	1	0.06	0.03	0.015	A742	K652, Y653, Y682, I706	NA	NA

(Continues)

TABLE 1 (Continued)

S. no.	Strain	Source	MIC (mg/L)			MIC			fks mutations			Antifungal use	Outcome
			FLU-MIC	AMB-MIC	CAS-MIC	ANI-MIC	MCF-MIC		fks1	fks2			
32	AKU-2020-183	Pus	64	0.25	0.12	0.12	0.015		A742	K652, I706		No	Alive
33	AKU-2019-482	Tissue	64	0.5	0.12	0.03	0.015		A742	W639		NA	NA
34	AKU-2020-192	Tissue	64	0.5	0.12	0.06	0.015		A742	T647, K652, I706		No	Alive
35	AKU-2019-177	Urine	32	0.25	0.03	0.12	0.06		A742	T647, K652, I706		FLU, VORI, AMB	NA
36	AKU-2019-408	Urine	128	0.5	0.12	-	-		A742	W639		NA	NA

Abbreviations: AMB, amphotericin; ANI, anidulafungin; CAS, caspofungin; FLU, fluconazole; MCF, micafungin; NA, information not available.

Note: Non-synonymous mutations in bold: higher caspofungin MICs isolates harbouring fks mutations. Synonymous mutation A2226T = A742, T1918C = W639, A1941T = T647, A1956G = K652, C1959T = Y653, T2046C = Y682, T2119C = I706. Synonymous mutation Y653 and Y682 were only found in blood and pus isolates.

isolates were initially pulverised using glass beads tube and homogenised at 4000rpm for 90s×3 = 0.05 (Precellys 24 lysis: Bertin Technologies). Extracted DNA was quantified using Nanodrop (DeNovixDS-11 FX+ Spectrophotometer/Fluorometer).

2.5 | PCR amplification and sequencing for *fks1* and *fks2* regions

The *fks1* and *fks2* genes regions were amplified using primers described previously.²⁷ Amplification reactions were set up in 25 µl reaction volume containing 12 µl master mix (BiobasicTaq Plus PCR master mix), 1 µl from each forward and reverse primer and 2 µl of 25 ng genomic DNA. The reaction cycling condition were: 2 min initial denaturation at 94°C, 35 cycles of 30 s at 94°C, 30 s at 56.7°C and 1 min at 72°C, and finally ending with 10 min at 72°C.

The PCR products were visualised on 2% agarose gels. Amplified products were purified using a PCR Purification Kit (QIAquick, Germany). Purified products were sequenced using sanger sequencing methodology with sequencing primers previously described.²⁷

2.6 | Sequence analysis

The *fks* gene sequences were analysed by MEGA6 Software. The nucleotide sequences of *fks1* and *fks2* were aligned and compared with the reference wild-type sequences of *C. glabrata* ATCC90030 *fks1* (GenBank accession numbers: HM366440.1) and *fks2* (GenBank accession numbers: HM366442.1), respectively. Point mutation in the amino acid sequence were identified by aligning the sequence using ClustalW (pairwise and multiple alignment). Non-synonymous mutations, i.e., change in the amino acid sequence of the protein, and synonymous mutation, i.e., change in the nucleotide that does not cause an amino acid change were identified.²⁸

3 | RESULTS

A total of 36 isolates were selected for the study. 11 isolates were from blood, 10 isolates from sterile fluid, 11 isolates from intra-abdominal pus, two isolates from deep tissue and two were from urine. The median (range) for caspofungin MIC was 0.06 (0.015–0.25) mg/L, micafungin MIC was 0.015 (0.008–0.06) mg/L and anidulafungin MIC was 0.06 (0.015–0.12) mg/L. For caspofungin, five isolates had MICs of 0.25 mg/L, 11 had MICs of 0.125 mg/L, 15 had MICs of 0.06 mg/L, four isolates had MICs of 0.03 mg/L and one had MIC of 0.015 mg/L (Table 1).

Hotspot region of *fks1* and *fks2* from 36 isolates was sequenced and analysed for the detection of mutations. *fks1* non-synonymous mutation D632H was observed in one isolate with caspofungin MIC of 0.25 mg/L. Synonymous mutation at position A742 was observed in 26/36 (72%) of the isolates (Table 1). For *fks2* mutation, 34 isolates were observed as wild type. A novel mutation at position

I661T was observed in *fk2* gene in one isolate with caspofungin MIC of 0.12 mg/L and anidulafungin and micafungin MIC of 0.06 and 0.015 mg/L, respectively. I661T was observed in proximity of S663P/F, a previously reported mutation that is associated with caspofungin resistance. Synonymous mutation in *fk2* gene at position W639 and Y682, T647, K652, Y653 and I706 were observed in 8/36 (22%), 16/36 (44%), 25/36 (69%), 6/36 (17%), and 23/36 (63%) isolates, respectively (Table 1).

Limited clinical data on treatment with echinocandins and outcome was available, hence, analysis of clinical outcomes with respect to in vitro results was not possible (Table 1).

4 | DISCUSSION

Our study reports an absence of phenotypic or genotypic resistance to echinocandins in a collection of clinical *C. glabrata* strains from the south of Pakistan. The tendency of *C. glabrata* to readily acquire resistance to major antifungal drug classes (azole and echinocandin) is a concern.²⁰ Mutations in *fk1* and *fk2* genes are the major mechanism of echinocandin resistance in *C. glabrata*. These mutations are differentially regulated and their expression levels may impact echinocandin resistance.²⁰ The phenotypic susceptibility testing among *C. glabrata* isolates failed to detect any resistance in our strains. However, the phenotypic susceptibility testing was performed on commercial colorimetric assay and reference method was not used, which is a limitation of our study. EUCAST recommends that isolates with high echinocandin MICs be tested for *fk* mutations to rule out testing artefact.²⁹ Pfaller et al.²⁴ reported that some isolates with *fk* mutations may have low echinocandin MICs and therefore, phenotypic in vitro testing is not 100% sensitive or specific in detecting echinocandin resistance.

Single-nucleotide polymorphism associated with echinocandin resistance in *fk1* gene at positions S629P/R/T and D632E/R/V/Y and *fk2* gene F659S/V/Y and S663F/P which are commonly reported globally, were not observed in this study.^{6,10,16,19,21,27,30–32} Among 36 isolates, novel single-nucleotide polymorphisms were detected in *fk* region in two isolates: D632H mutation in *fk1* in one and I661T in *fk2* gene in another. Both patients with *C. glabrata* strains with non-synonymous mutations at *fk1* and *fk2*, were not treated with echinocandins. Hence, no conclusions on therapeutic response can be made.

Seven synonymous mutations were identified. Of which, six were observed in *fk2* gene. Synonymous mutations in *fk* were detected in strains with higher caspofungin MICs (0.03–0.25 µg/ml). Twenty-three different amino acid substitutions in *fk1* and *fk2* region as previously described, S629P and D632E/G in *fk1*, and F659V/Y and S663P in *fk2* are the most reported mutations.^{6,19,27,32} A study from France reported no *fk1* mutations, however, two mutations, F659 and F708V, were observed in *fk2* gene.³³ Mutation in *fk1* or *fk2* have been associated with elevated echinocandin MICs, reduced glucan synthase enzyme sensitivity and any prior echinocandin exposure.³²

Echinocandin treatment failure has also been observed in patients with *C. glabrata* isolates having low echinocandin MICs and mutation in *fk* gene.^{15,16} Shields et al.¹⁵ found that detection of *C. glabrata* *fk* mutation was better than MICs in predicting clinical failure of therapy among patients with invasive candidiasis. They reported that therapeutic decisions based on *fk* mutation in *C. glabrata* isolate in a patient and change in therapy to amphotericin led to improved treatment response.¹⁵ However, no such conclusion could be drawn from our finding since only one patient received caspofungin as treatment in combination with amphotericin and expired after 12 days of therapy. The *C. glabrata* isolated from this patient had synonymous mutation in *fk1* and *fk2*. Information on treatment and outcome was not available for patient with *C. glabrata* non-synonymous mutation in *fk2* (I661T), but it had higher MIC only to caspofungin; while the patient with *fk1* (D632H) mutation-bearing strain had no prior history of antifungal use and caspofungin MIC was also within susceptible range.

Low frequencies of both non-synonymous and synonymous polymorphisms were observed in invasive *C. glabrata* strains in this study. S663P in *fk2*-gene has been reported to be associated with caspofungin resistance. A small number of non-synonymous mutations in this study, those too with doubtful relevance to clinical resistance, identifies need to conduct large-scale periodic genomic surveillance and correlate with treatment outcome data. In India and Iran, no *fk* mutation nor phenotypic resistance were observed in *C. glabrata* isolates, supporting our findings.^{11,34,35} However, other studies from Central Asia report presence of S663P and ΔF659 in phenotypically echinocandin resistant *C. glabrata* isolates.¹⁹

This study reports the rate and position of synonymous and non-synonymous mutations prevalent in local *C. glabrata* isolates. Nevertheless, the study has limitations because of small sample size and it was a single-centre study, with an exceedingly small proportion of isolates from outside Karachi. We did not use CLSI reference method for performing MIC testing and colorimetric methods may have given an error of 1–2 dilutions compared with reference, although categorical agreement was very high when testing all three echinocandins.³⁶ There was incomplete treatment outcome data to draw any conclusion regarding the mutations identified. A prospective outcome of response to echinocandin treatment in the patients would have given substantial weight to our findings.

The study shows that genotypic profiling together with antifungal susceptibility could provide epidemiological data to inform regional treatment recommendations for patients with *C. glabrata* candidemia.

AUTHOR CONTRIBUTIONS

Study design: Kauser Jabeen, Urooj Zafar, Saba Memon, Najia Karim Ghanchi, Joveria Farooqi. Protocol development and methodology: Kauser Jabeen, Urooj Zafar, Saba Memon, Najia Karim Ghanchi, Joveria Farooqi. Collection of data: Joveria Farooqi, Saba Memon, Sadaf Zaka. Data analysis and interpretation: Najia Karim Ghanchi, Saba Memon. Manuscript writing initial draft: Kauser Jabeen, Saba Memon. Manuscript review and approval: All authors.

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CONFLICT OF INTEREST

None for any of the authors.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

CONSENT FOR PUBLICATION

All authors have seen and approved the final version of this manuscript and given their consent for publication in this journal.

ORCID

Kauser Jabeen  <https://orcid.org/0000-0003-2497-7847>

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